Prodigy Anesthesia Quick Review

Hemophilia

What is hemophilia A?	Hemophilia A is an X-linked recessive clotting disorder that results in a deficiency of factor VIII. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 421-422.*
What are the most common presenting symptoms of hemophilia A?	Patients with hemophilia A are usually diagnosed in childhood with spontaneous bleeding into joints and vital organs. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 421.*
How does the severity of hemophilia A vary and how can the severity be assessed?	Severe hemophilia A exists when the factor VIII concentration is less than 1% of normal. Patients whose factor VIII levels are between 6% and 30% of normal may not suffer spontaneous bleeding or hemarthroses, but are at increased risk of bleeding during surgery. Even female carriers can have a mild form that can reduce factor VIII activity to 30% of normal. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 421.*
How are coagulation profiles affected by hemophilia A?	The PTT is significantly prolonged in severe hemophilia A and only moderately prolonged in milder forms of hemophilia A. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 421.*
What is the treatment for hemohilia B?	Factor IX concentrate. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 422.*
What is the treatment for hemophilia A?	Factor VIII concentrate *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 421.*
What preparations should be made for the patient with hemophilia A about to undergo surgery?	Factor VIII concentrate should be administered until levels are at or near 100% of normal prior to surgery, which may require a dose of 50-60 units/kilogram. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 421.*
What is the half-life of factor VIII and what is the significance of this?	The half-life of factor VIII is about 12 hours, which means that during the perioperative period, infusions of 30 units/kg may have to be repeated every 8-12 hours to keep factor VIII levels at half of normal. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 421.*

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Hemophilia

What is hemophilia B?	Hemophilia B is an X-linked recessive coagulation disorder caused by a deficiency of factor IX. Because factor IX is also known as Christmas factor, hemophilia B is sometimes known as Christmas disease. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 422.*
How does the severity, course of illness, and laboratory findings in hemophilia B compare to that of hemophilia A?	Except for the specific clotting factor involved, the course of illness with hemophilia B is almost identical to that of hemophilia A. Also like hemophilia A, severe hemophilia B occurs when less than 1% of the normal amount of factor is present and becomes considerably milder when patients have between 5% and 40% of the normal amount of factor. Like hemophilia A, patients with hemophilia B exhibit a prolonged PTT and a normal PT. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 422.*
What are the two treatment regimens available for hemophilia B?	The two main treatments available are recombinant factor IX and factor IX-PCC preparation. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 422.*
What is Factor XI deficiency?	Factor XI deficiency is known as hemophilia C and occurs in approximately 4% of Ashkenazi Jews. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 422.*
What is Alexander's disease?	Alexander's disease is a congenital factor VII deficiency. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 422.*
Why would one choose to give recombinant factor IX over factor IX-PCC?	Factor IX-PCC contains a high concentration of activated clotting factors. When enough is given to raise the factor IX level to 50% of normal, a significant risk of thromboembolism develops, especially in the setting of orthopedic surgery. If only factor IX is needed, then the administration of Factor IX-PCC may be an unnecessary risk. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 422.*
How does the dosing regimen for factor IX compare to that of factor VIII?	Like factor VIII treatment, factor IX must be administered over several days during the perioperative period, but because collagen can absorb factor IX, the doses that must be administered are about double that of factor VIII. It has a longer half-life than factor VIII, however, doses of 100 units/kilogram every 12 to 24 hours is suitable to keep factor IX levels above 50%. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 421-422.*

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